

The Best Open Source Machine Learning and Deep Learning Tools for Data Science

Deriving knowledge and insights from data empowers enterprises to make effective decisions, cut costs, ensure quick delivery, as well as enhance competitiveness and productivity. Enterprises that use data science as an integral part of their business strategy score over businesses that ignore it.



Data science is regarded as the systematic study and analysis of data from various sources for effective decision making and problem solving. The objective of data science is to facilitate the application of various processes related to data like data procurement, data pre-processing for noise clearing, data representation, data evaluation, data analysis and the use of data to create knowledge related to the enterprise. Data science contributes in a significant manner towards decision making at varying levels—individual, organisational and global. The ultimate emerging opportunity in data science is Big Data, which enables organisations to analyse a large number of data sets generated through website logs, sensors

and data transactions to gain insights and make effective decisions.

The term 'data science' has appeared in several contexts since the past 30 years and has recently gained significant attention. An early reference to data science was by Peter Naur in 1960 when a term titled 'atalogy' was coined. In 1974, Naur published his book, 'A Concise Survey of Computer Methods', which freely used the term data science in its survey of the contemporary data processing methods being used in a wide range of applications.

The importance of data science and analytics in enterprises

In the modern era, data is regarded as a key element to run any enterprise

successfully. It is not enough for the business to just acquire data, but to attain better results, data should be effectively used. Data must be processed and analysed in proper ways to gain insights into a particular business. When data is acquired from multiple sources, with no specific format, it incorporates lots of noise and redundant values. In order to make it suitable for decision making, data has to undergo the processes of cleaning, munging, analysing and modelling. This is where data science and analytics come into play. Big Data analytics has been a major revolution for enterprises, since it has strengthened the roots of businesses. As of 2018, more than 50 per cent of the world's enterprises are